

CYBR-490 Capstone Project

Milestone 1 Report

Project Title:

AI Threat Intelligence Aggregator

Team Members:

Cassandra, Tamim, Monirul

Instructor:

Yanyan Li

Date:

03/10/2026

Changes to the Plan Document

Since the submission of the initial project plan, several updates have been made to the system design and development approach. These changes reflect progress in implementation and refinement of the system architecture.

The primary updates include:

1. Integration of Threat Intelligence Sources

The system now incorporates two initial threat intelligence sources:

- **CISA Known Exploited Vulnerabilities (KEV) Catalog**
- **MITRE ATLAS AI Threat Knowledge Base**
- **AI Security Research Papers retrieved through Semantic Scholar**

These sources were selected because they provide high-quality security intelligence and relevant information related to emerging threats in both traditional cybersecurity systems and artificial intelligence environments.

2. Development of Data Ingestion Pipeline

The architecture has been updated to include a Python-based data ingestion pipeline responsible for retrieving and parsing threat intelligence data from external sources.

The ingestion layer standardizes data collected from different formats and prepares it for processing within the system.

3. Structured JSON Threat Dataset

Threat information is now stored in a **structured JSON format**. This format enables efficient processing and supports future integration with AI models and the dashboard interface.

The JSON dataset acts as the central data structure connecting the ingestion pipeline and the visualization layer.

4. Threat Categorization Framework

The project plan has been updated to include a threat classification structure consisting of three categories:

- **Hypothetical Threats**
- **Demonstrated Threats**
- **Active Exploitation**

This categorization helps security analysts better understand the maturity and severity of AI-related threats.

Updated Design Artifacts

System Requirements

The following requirements guide the development of the AI Threat Intelligence Aggregator.

Functional Requirements

The system must:

1. Collect threat intelligence data from external sources.
2. Parse and normalize threat information into a structured format.
3. Store threat information in a standardized JSON dataset.
4. Categorize threats into predefined threat groups.
5. Display threat information through a graphical dashboard.
6. Allow users to explore and filter threats.

Non-Functional Requirements

The system should:

- maintain reliable data processing pipelines
- support scalable threat ingestion
- ensure data integrity and validation
- present threat intelligence in a user-friendly interface

System Architecture

The system architecture consists of several layers responsible for collecting, processing, storing, and displaying threat intelligence data.

Architecture Overview

Threat Sources
(CISA KEV / MITRE ATLAS / Research Papers)

|
v

Data Ingestion Layer
(cisa_kev_ingest.py
mitre_atlas_ingest.py
semantic_scholar.py)

|
v

Data Processing Pipeline
(cleaning, normalization, schema mapping)

|
v

AI Processing Layer (Research Papers)
(ai_agent.py - LLM summarization)

|
v

Structured JSON Datasets
(cisa_kev_data.json
atlas_data.json
paper_summaries.json)

|
v

Dashboard GUI
(gui.py - threat visualization interface)

Implementation Progress

At this stage of development, several core components of the system have been implemented or are currently under development.

Data Ingestion

The project currently includes ingestion mechanisms for the following threat intelligence sources:

- **CISA KEV Catalog**
- **MITRE ATLAS database**
- **AI Security Research Papers retrieved through Semantic Scholar**

Python scripts retrieve relevant threat information and extract key fields from these sources. Research papers are processed through an additional pipeline that retrieves paper metadata and uses an AI agent to generate summarized threat insights.

Data Processing Pipeline

The processing pipeline performs several tasks:

- cleaning raw data
- normalizing threat fields
- structuring records into a consistent schema

This pipeline ensures that data from different sources can be processed consistently.

JSON Threat Data Structure

Threat records are stored in a structured JSON format. Each record contains information describing a threat entry.

Example JSON record:

```
</> JSON

{
  "title": "Prompt Injection Attack",
  "source": "Research Paper",
  "summary": "Attack manipulating LLM prompts to influence model output",
  "category": "Demonstrated",
  "risk_level": null,
  "url": "https://example-source"
}
```

Updated Timeline

The following timeline summarizes completed tasks and remaining work.

Tasks Completed or In Progress

Task	Status
Data ingestion framework	In Progress
Data processing pipeline	In Progress
CISA KEV integration	In Progress
MITRE ATLAS integration	In Progress
Research paper dataset ingestion	In Progress

Next Development Phase

The next stage of development will focus on completing the integration between the backend data pipeline and the graphical dashboard interface.

Key objectives include:

- finalizing threat categorization
- improving the dashboard interface
- integrating Python modules into a unified application
- improving project organization and code structure

Conclusion

This milestone represents significant progress toward building the AI Threat Intelligence Aggregator. The development team has established the core data ingestion and processing pipeline and has begun integrating real-world threat intelligence sources.

Future work will focus on completing system integration, improving the dashboard interface, and expanding the threat dataset to support advanced threat analysis.